

$$\begin{array}{c} \alpha \\ \alpha \\ \alpha \\ \alpha \\ \alpha \end{array} = (1 - e^{4i\alpha}) \begin{array}{c} \alpha \\ \pi \end{array} + 2e^{2i\alpha}(e^{2i\alpha} + 1) \begin{array}{c} \alpha \\ \pi \end{array} + 2e^{2i\alpha}(e^{2i\alpha} - 1) \begin{array}{c} \alpha \\ -\frac{\pi}{2} \\ \pi \end{array} \begin{array}{c} \frac{\pi}{2} \\ \frac{\pi}{2} \\ \frac{\pi}{2} \\ \frac{\pi}{2} \\ \frac{\pi}{2} \end{array}$$

Diagram illustrating a mathematical identity involving complex exponentials and a sum of terms. The left side shows a vertical stack of five green circles, each labeled α , with a horizontal line extending to the right from each circle. This is followed by an equals sign and three terms separated by plus signs.

The first term is $(1 - e^{4i\alpha})$ multiplied by a diagram consisting of a green circle labeled α connected to a green circle labeled π . The π circle has five curved lines extending to the right.

The second term is $+ 2e^{2i\alpha}(e^{2i\alpha} + 1)$ multiplied by a diagram consisting of a green circle labeled α connected to a red circle labeled π . The π circle has five curved lines extending to the right.

The third term is $+ 2e^{2i\alpha}(e^{2i\alpha} - 1)$ multiplied by a diagram consisting of a green circle labeled α connected to a green circle labeled $-\frac{\pi}{2}$, which is then connected to a red circle labeled π . The π circle has five curved lines extending to the right, each ending in a green circle labeled $\frac{\pi}{2}$.